r/NoStupidQuestions

What happens to AI used water?

Posted by u/MicroSnaked • • 0 points • 15 comments

So I watched a video on how AI uses water, and that it's basically recycled a bunch of times and then it's evaporated, right (correct me if Im wrong)? But when people say things along the lines of "AI is bad for the environment because uses so much water," wont it get recycled again? I read that AI roughly uses 0.32 milliliters per prompt, but where does that 0.32 go? In other words, does that 0.32 get used multiple times for different prompts kind of? Is it just gone forever?

TL;DR: How bad is AI water usage if it just gets recycled?

Comments

u/Strange_Bank6779 • 1 points • 2026-03-03 11:19 UTC

You're not allowed to ask these kinds of questions. It undermines the climate narrative.

u/Less_Ability_5721 • 1 points • 2026-03-02 03:09 UTC

In big data centers, heat is a huge issue. Computers produce a ton of heat, which you have to get rid of in order for the computers to keep working.

They'll likely have some sort of cooling system that, eventually, transfers as much heat as possible to water. In order to reuse the water, you have to cool it off.

A cheap and effective way to cool off water is to use evaporative cooling. You encourage some of the water to evaporate, and that cools the rest of it off.

You know how, if you're sweating, you'll feel a lot colder if you stand right by a fan? It's that concept, but on an industrial scale.

You'll have a giant cooling tower, and the hot water gets showered down from the top while huge fans draw air through it. A bit of the water evaporates, which cools the rest off, and the much cooler water collects at the bottom and is recycled back into the cooling system.

So, here's how an AI query uses water.

You ask an AI something, and it uses a lot of power to think of an answer. That power generates a lot of waste heat. The building's AC unit pumps some water through its system while it cools the room down That water gets taken outside and runs through a cooling tower, where 99% is cooled off and is able to be reused on the next query, but 1% is lost as steam.

That steam will eventually condense into clouds and rain down somewhere, but that doesn't do any immediate good to the general vicinity, and AI data centers are not always built in areas with abundant sources of fresh, clean water. Central Texas has become a huge hub for AI data centers, but they've had several years in a row of drought conditions. And, honestly, even in the best of years, they're a relatively arid area that doesn't have a ton of water to spare.

Water is a renewable resource, but renewing it takes time, and when you use it faster than it can renew, you're causing huge issues.

u/sdn • 1 points • 2026-03-02 02:16 UTC

I live in San Antonio, TX which gets it water from the [Edwards Aquifer](https://en.wikipedia.org/wiki/Edwards_Aquifer). Any time it rains north of San Antonio, water percolates into the ground and then after months eventually makes it down to the level of the aquifer.

For the past \~5 years or so, we've been in a drought here. The aquifer is in stage 4 water conservation (stage 5 is the worst). It hasn't rained north of the city here enough to replenish the aquifer in many years.

So you ask - doesn't the water get recycled?

Well, sort of - eventually - on a long enough time scale. When I flush my toilet in San Antonio, the water goes into the sewer and then eventually into a settling pond south of town. Once the water has been filtered, it gets dumped into the San Antonio River and eventually that water makes it into the Gulf of Mexico.

If the weather patterns are right, that water gets evaporated out of the gulf and eventually falls down somewhere - hopefully on San Antonio.

Whenever someone builds a data center in San Antonio, then they need to cool the equipment. Water cooling is very efficient for doing that. So what do you do? You pump cool water out of the aquifer, run it through a heat exchanger which has cool the hot glycol coming off the equipment, then you dump the hot water into the sewer.

So.. that's how AI is "using" water. The amount of water on the planet stays the same, but clean drinking water slowly gets reduced.

u/ElkEquivalent4625 • 1 points • 2026-03-02 01:57 UTC

Heavy groundwater extraction worsens local water scarcity

u/DeadGuyInRoom4 • 1 points • 2026-03-02 01:56 UTC

It reduces the availability of fresh drinking water in areas where fresh water is already a strained resource.

u/Klaargs_ugly_stepdad • 3 points • 2026-03-02 01:49 UTC

We have a limited amount of fresh water. Water cooling requires fairly pure fresh water. If it's stuck in a cooling system, that amount is taken away from the total amount of water we have, and it isn't available for anyone to drink.

Also, if it's just evaporated off, it goes into the sky and the wind may or may not blow it somewhere that the rain re-enters the water table it was taken from. This is worse than when you flush the water back into the sewers for treatment, because that way it definitely stays in the area it was taken from.

u/Wrong-Election1997 • 1 points • 2026-03-02 01:40 UTC

Water usage is tied to energy consumption

u/inorite234 • 7 points • 2026-03-02 01:40 UTC

When they say "water usage is bad for the environment" they mean clean, fresh water usage.

I can pull whatever the fuck water I want from every source available.....but AI datacenters aren't using dirty ass, Crick water. They're pulling clean, fresh water. That water is either a source for drinking purification facilities, or it's already been through the purification process. That process takes time and energy.

u/KronusIV • 2 points • 2026-03-02 01:36 UTC

A lot of the water is evaporated, and a lot goes down the drain. Either way, it's of no use to the local community any more. Sure, it'll eventually go through the water cycle and rain down somewhere, but that's likely to be hundreds or thousands of miles away from where it was used. That's the big issue.

> **u/Minimum_Run_890** • 1 points • 2026-03-02 01:40 UTC
>
> And it won’t happen tomorrow either

u/anti-beep • 9 points • 2026-03-02 01:35 UTC

Yes, eventually it will fall back down, that is correct. But.. it wont fall down in the same place it came from.

If the basins containing the water are drained faster than they replenish.. that’s gonna ruin the local ecosystems.

> **u/LowGravitasIndeed** • 1 points • 2026-03-03 07:36 UTC
>
> This. The majority of rainwater on Earth falls into the sea and isn't reclaimable without desalination (which poses multiple other issues).
> There's a lot of water on Earth, relatively little of it is readily usable though, and we spend a lot of energy purifying and distributing it. Wasting any of it is bad, mkay?

u/eggs-benedryl • 0 points • 2026-03-02 01:33 UTC

Yes the water cycle continues.

u/hellshot8 • 4 points • 2026-03-02 01:31 UTC

the issue is how much energy it uses. the water thing is sortof a misnomer

> **u/KronusIV** • 7 points • 2026-03-02 01:37 UTC
>
> Nah, both are big problems. One might be more urgent than the other, it depends on the region.